

SYSTEM ANALYSIS AND DESIGN

SEGMENT-5

Acknowledgments

The contents of this slide has been mostly prepared by the course materials integration of respectable course teachers ***Abu Nowhash Chowdhury***[Junior Instructor of Computer Science, Pre-Undergraduate Program, Asian University for Women & Adjunct Faculty, IIUC] and ***Mr. Sudipta Vaskar Rakshit*** [Deputy Manager, (ICT-Software), Pacific Jeans Ltd. & Adjunct Faculty, IIUC]. I would like to thank sincerely for their cordial co-operation.

Process of design

The design process

Think of any digital experience or product you've enjoyed in your everyday life, like scrolling Instagram or TikTok for hours on end, reading Reddit on the toilet, looking up travel destinations on Google, or ordering groceries through Amazon or Instacart. The list goes on and on. Each of these digital products have countless features that are thoughtfully designed and highly performant — but that wasn't always the case.

Every part of those products, down to the corner radius of a button, started out as a concept. From there, each concept journeyed through a design process that resulted in what you see and experience today.

In this article, we'll talk about what the design process looks like and why it's important for design teams to create a process that nurtures the best of their creative ideas from concept to final design.

The design process (and why it is important)

Achieving good design takes more than simply time and talent. It takes clarity around a specific goal, and when a larger team is involved, collaboration and alignment. For that reason, the design process is crucial because it's the disciplined approach that guides creative teams towards a clear and meaningful goal for their design work.

The design process is very important to non-designers, too. For example, when you're designing a digital product, cross-functional team members such as product managers, researchers, analysts, developers, and other stakeholders have valuable input for what is being designed. The design process makes it easier for those different individuals to get involved at the right time, such as during market research, brainstorming, user testing, and Design Delivery.

Ultimately, the design process provides healthy guidelines to ensure that your design concepts meet user needs while staying brand consistent.

What does the design process look like?

No two creatives have the same exact process and that's also true for design teams and their design processes. Teams and individual designers who are focused on UI design will have a different approach than those who are doing graphic design or interaction design.

That said, there is a common set of steps that most teams follow when taking on any substantial design project.

Design process steps

The five main steps in the design process are Empathize, Define, Ideate, Deliver, and Test. You might hear different names to describe these steps, but this is how most design processes unfold. And as for what happens within each step, it depends on what you're designing, your team setup, and other constraints in your organization such as budget, industry, and more.

Step 1: Empathize

The first step in the design process, also referred to as "Discovery", involves the same initial effort required for any new business challenge: identifying and empathizing with your target user and their needs. This effort to deeply understand your users includes activities like market research, UX research, conducting user interviews, and finally, bringing together all of the information you've discovered so that it contributes to and improves your design thinking.

This step takes longer when you're designing a brand new product or if your design project is complex, like an application, website, or comprehensive digital experience, because the amount of user research you'd need to fully empathize with your end user is greater.

Step 2: Define

What have you learned about your user? What problems are they running into? How are they currently dealing with them? What could you create to make their life better?

These are some questions to answer at the second step of the design process, which is all about defining your problems, goals and boundaries. Here, you'd take what you've learned about your users' needs and narrow down what matters. For some teams, this step might include creating user flows and personas, drafting design briefs, and holding meetings to gain alignment on design scope, deliverables, responsibilities, and timelines.

Step 3: Ideate

The third step of the design process is what most people envision when they think of a designer's creative process. Often messy to some degree, the ideate stage is where designers brainstorm ideas, challenge assumptions, and ultimately create a solution to a problem they've defined in earlier steps. Typically, the outcome of this step is some sort of design work product, such as wireframes, mockups, or first drafts.

Step 4: Deliver

The deliver step of the design process is the turning point when designs begin to feel real because they are shared with others in the form of a prototype or MVP. When building digital products, this is where the importance of design delivery comes into play because this determines how

efficiently developers can transform the design vision into a final outcome that solves users' needs.

In this step, you'll often find design presentations, design handoff meetings, and depending on your design tool of choice, Figma, Sketch, XD and Photoshop.

Step 5: Test

When designing digital products and experiences today, you'll rarely find a designer designing something and never touching it again. That's because the most successful designs are a result of an iterative approach, where there's a continuous cycle of updates and improvements to the design based on feedback from users and customers. After a design has been user tested, it may go back to the ideate stage to get further refined.

The importance of the design process

The design process provides structure to what might otherwise be lots of creative ideas that have nowhere to go. Establishing a design process improves team collaboration, increases the quality and speed of your design work, and makes it more likely that everyone – especially your end users – is happy with your final designs.

Here are some specific positive benefits of a solid design process:

1. Set clearer expectations

While an individual designer's creative process might be somewhat chaotic and messy, the design process as a whole is not. There's a predictable sequence of activities that everyone in a larger organization can understand. As a result, timelines are more predictable and there's less need to wonder "what's happening with the design?"

2. Improve teamwork and alignment

You can have the most talented design team in the world, but it doesn't matter if there isn't clarity and a common workflow that gets everyone working in sync. Maintaining an organized design process leads to better teamwork and alignment by providing clarity on design scope and objectives, who is responsible for what, and which activities must happen in what order.

3. Create more efficiently

The design process is like a workflow template to follow, and we all know that templates help us all work faster. So with everyone on the same page, multidisciplinary team members know how and when they can contribute during the design process, which means you can make important design decisions more quickly along the way.

4. Deliver final designs that meet user and business needs

The design process forces you to consider your user and their needs from the very start, which reflects positively in your final design and user experience. And because the design process also gives space to proper delivery and testing, your team benefits from less friction during developer handoff and more certainty that the final designs meet all of the goals you intended for them.

5. Gain happy users

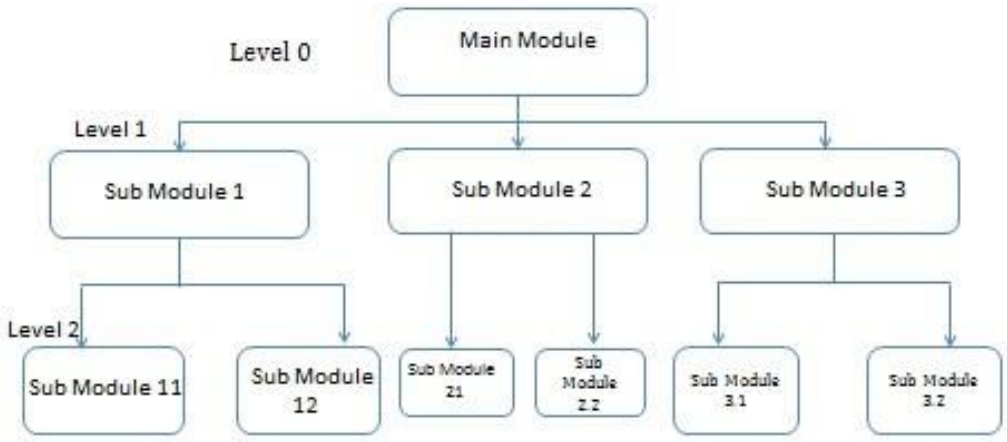
The ultimate goal for any design project is not just to solve a problem, but to also make users happy. The design process gives you that extra layer of assurance that you're doing the right thing and creating something that your users will actually enjoy.

Design Methodologies

Top-Down Strategy

The top-down strategy uses the modular approach to develop the design of a system. It is called so because it starts from the top or the highest-level module and moves towards the lowest level modules.

In this technique, the highest-level module or main module for developing the software is identified. The main module is divided into several smaller and simpler submodules or segments based on the task performed by each module. Then, each submodule is further subdivided into several submodules of next lower level. This process of dividing each module into several submodules continues until the lowest level modules, which cannot be further subdivided, are not identified.

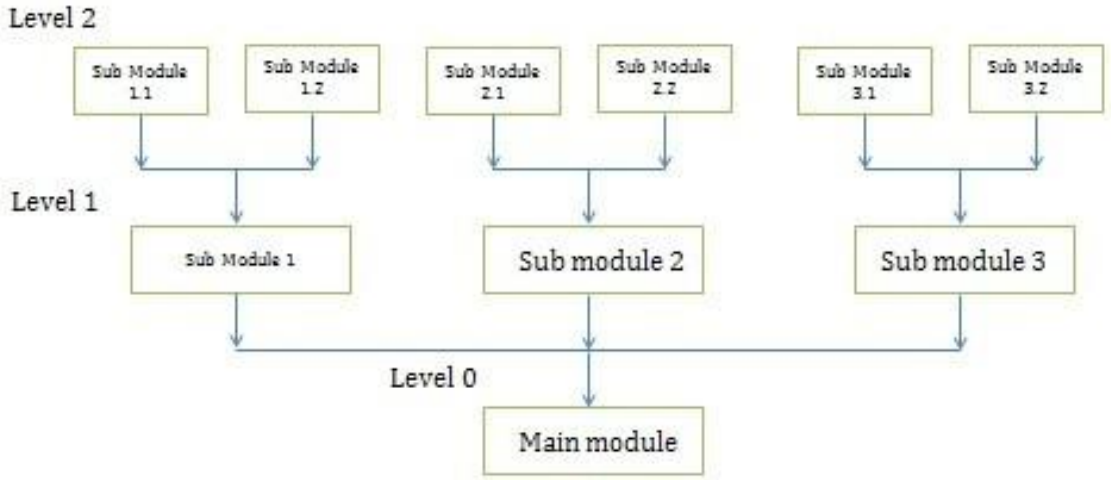


Bottom-Up Strategy

Bottom-Up Strategy follows the modular approach to develop the design of the system. It is called so because it starts from the bottom or the most basic level modules and moves towards the highest level modules.

In this technique,

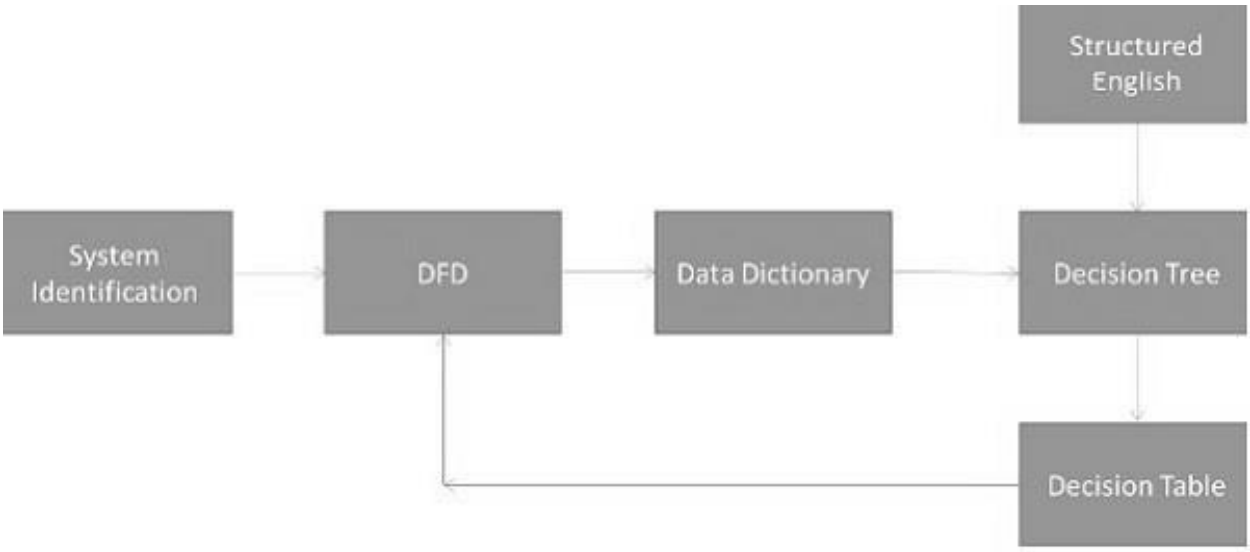
- The modules at the most basic or the lowest level are identified.
- These modules are then grouped together based on the function performed by each module to form the next higher-level modules.
- Then, these modules are further combined to form the next higher-level modules.
- This process of grouping several simpler modules to form higher level modules continues until the main module of system development process is achieved.



Structured Design

Structured design is a data-flow based methodology that helps in identifying the input and output of the developing system. The main objective of structured design is to minimize the complexity and increase the modularity of a program. Structured design also helps in describing the functional aspects of the system.

In structured designing, the system specifications act as a basis for graphically representing the flow of data and sequence of processes involved in a software development with the help of DFDs. After developing the DFDs for the software system, the next step is to develop the structure chart.



Modularization

Structured design partitions the program into small and independent modules. These are organized in top down manner with the details shown in bottom.
Thus, structured design uses an approach called Modularization or decomposition to minimize the complexity and to manage the problem by subdividing it into smaller segments.
Advantages

- Critical interfaces are tested first.
- It provide abstraction.
- It allows multiple programmers to work simultaneously.
- It allows code reuse.
- It provides control and improves morale.
- It makes identifying structure easier.

Input Design

In an information system, input is the raw data that is processed to produce output. During the input design, the developers must consider the input devices such as PC, MICR, OMR, etc.

Therefore, the quality of system input determines the quality of system output. Well-designed input forms and screens have following properties −

- It should serve specific purpose effectively such as storing, recording, and retrieving the information.
- It ensures proper completion with accuracy.
- It should be easy to fill and straightforward.
- It should focus on user's attention, consistency, and simplicity.
- All these objectives are obtained using the knowledge of basic design principles regarding

–

  - What are the inputs needed for the system?
  - How end users respond to different elements of forms and screens.

Objectives for Input Design

The objectives of input design are −

- To design data entry and input procedures
- To reduce input volume
- To design source documents for data capture or devise other data capture methods
- To design input data records, data entry screens, user interface screens, etc.
- To use validation checks and develop effective input controls.

Data Input Methods

It is important to design appropriate data input methods to prevent errors while entering data. These methods depend on whether the data is entered by customers in forms manually and later entered by data entry operators, or data is directly entered by users on the PCs.

A system should prevent user from making mistakes by −

- Clear form design by leaving enough space for writing legibly.
- Clear instructions to fill form.
- Clear form design.
- Reducing key strokes.
- Immediate error feedback.

Some of the popular data input methods are −

- Batch input method (Offline data input method)
- Online data input method
- Computer readable forms
- Interactive data input

Output Design

The design of output is the most important task of any system. During output design, developers identify the type of outputs needed, and consider the necessary output controls and prototype report layouts.

Objectives of Output Design

The objectives of input design are −

- To develop output design that serves the intended purpose and eliminates the production of unwanted output.
- To develop the output design that meets the end users requirements.
- To deliver the appropriate quantity of output.
- To form the output in appropriate format and direct it to the right person.
- To make the output available on time for making good decisions.

Designing Effective Output

Output Method	Advantages	Disadvantages
Printer	• Affordable for most organizations • Flexible in types of output, location, and capabilities • Handles large volumes of output • Highly reliable with little down time	• Still requires some operator intervention • Compatibility problems with computer software • May require special, expensive supplies • Depending on model, may be slow • Environmentally unfriendly
Display screen	• Interactive • Online, real-time transmission • Quiet • Takes advantage of computer capabilities for movement within databases and files • Good for frequently accessed, ephemeral messages	• May require cabling and setup space • Still may require printed documentation
Audio output and podcasts	• Good for individual user • Good for transient messages • Good where worker needs hands free • Good if output needs to be widely distributed	• Needs earbuds where output will not interfere with other tasks • Has limited application
DVD, CD-ROM, and CD-RW	• Has large capacity • Allows multimedia output	• Requires a computer and display for reading data
Electronic output (email, Web sites, blogs, and RSS feeds)	• Reduces paper • Can be updated very easily • Can be "broadcast" • Can be made interactive	• Is not conducive to formatting (email) • Is difficult to convey context of messages (email) • Web sites need diligent maintenance

Nature of Output Reports

- **TOP MANAGEMENT**

  - Summary highlighting important results

  Graphical Output – Pie charts

  – Bar charts

  – Maps

  Needed for strategic management

Let us now go through various types of outputs –

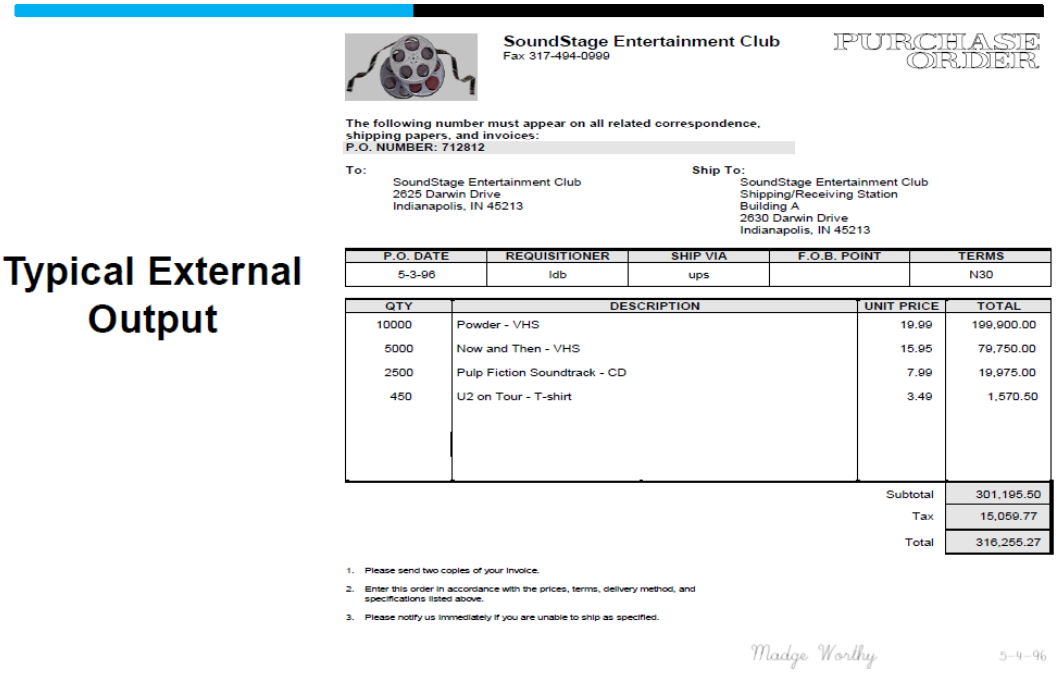
External Outputs

Manufacturers create and design external outputs for printers. External outputs enable the system to leave the trigger actions on the part of their recipients or confirm actions to their recipients.

Some of the external outputs are designed as turnaround outputs, which are implemented as a form and re-enter the system as an input.

- **Turnaround outputs** are those which are typically implemented as a report eventually re-enters the system as an input

Typical External Output



SoundStage Entertainment Club
Fax 317-494-0999

PURCHASE ORDER

The following number must appear on all related correspondence, shipping papers, and invoices:
P.O. NUMBER: 712812

To:
SoundStage Entertainment Club
2625 Darwin Drive
Indianapolis, IN 45213

Ship To:
SoundStage Entertainment Club
Shipping/Receiving Station
Building A
2630 Darwin Drive
Indianapolis, IN 45213

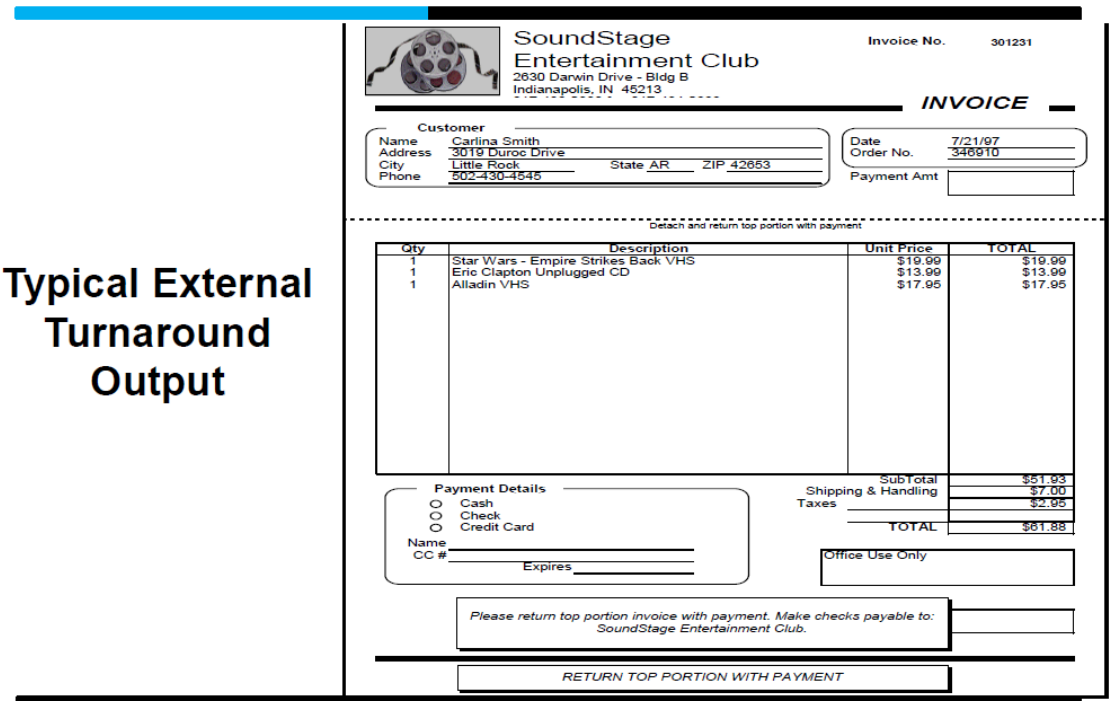
P.O. DATE	REQUISITIONER	SHIP VIA	F.O.B. POINT	TERMS
5-3-96	ldb	ups		N30

QTY	DESCRIPTION	UNIT PRICE	TOTAL
10000	Powder - VHS	19.99	199,900.00
5000	Now and Then - VHS	15.95	79,750.00
2500	Pulp Fiction Soundtrack - CD	7.99	19,975.00
450	U2 on Tour - T-shirt	3.49	1,570.50
		Subtotal	301,195.50
		Tax	15,059.77
		Total	316,255.27

1. Please send two copies of your invoice.
2. Enter this order in accordance with the prices, terms, delivery method, and specifications listed above.
3. Please notify us immediately if you are unable to ship as specified.

Madge Worthy 5-4-96

Typical External Turnaround Output



SoundStage Entertainment Club
2630 Darwin Drive - Bldg B
Indianapolis, IN 45213

Invoice No. 301231

INVOICE

Customer
Name Carlina Smith
Address 3019 Duroc Drive
City Little Rock State AR ZIP 42653
Phone 502-430-4545

Date 7/21/97
Order No. 348910
Payment Amt

Detach and return top portion with payment

Qty	Description	Unit Price	TOTAL
1	Star Wars - Empire Strikes Back VHS	$19.99	$19.99
1	Eric Clapton Unplugged CD	$13.99	$13.99
1	Alladin VHS	$17.95	$17.95

Payment Details
○ Cash
○ Check
○ Credit Card
Name
CC #
Expires

SubTotal $51.93
Shipping & Handling $7.00
Taxes $2.95
TOTAL $61.88

Office Use Only

Please return top portion invoice with payment. Make checks payable to: SoundStage Entertainment Club.

RETURN TOP PORTION WITH PAYMENT

Internal outputs

Internal outputs are present inside the system, and used by end-users and managers. They support the management in decision making and reporting.

There are three types of reports produced by management information −

- **Detailed Reports** − They contain present information which has almost no filtering or restriction generated to assist management planning and control.

    - ✓ Some detailed reports are historic in nature
    - ✓ Detailed reports serve as an audit trail for subsequent management inquiries

- **Summary Reports** − They contain trends and potential problems which are categorized and summarized that are generated for managers who do not want details.
- **Exception Reports** − They contain exceptions, filtered data to some condition or standard before presenting it to the manager, as information.
    - ✓ Exception reports only report exceptions to some condition or standard

Forms Design

Both forms and reports are the product of input and output design and are business document consisting of specified data. The main difference is that forms provide fields for data input but reports are purely used for reading. For example, order forms, employment and credit application, etc.

- During form designing, the designers should know –
  -who will use them
  -where would they be delivered
  -the purpose of the form or report

- During form design, automated design tools enhance the developer's ability to prototype forms and reports and present them to end users for evaluation.

Objectives of Good Form Design

A good form design is necessary to ensure the following −

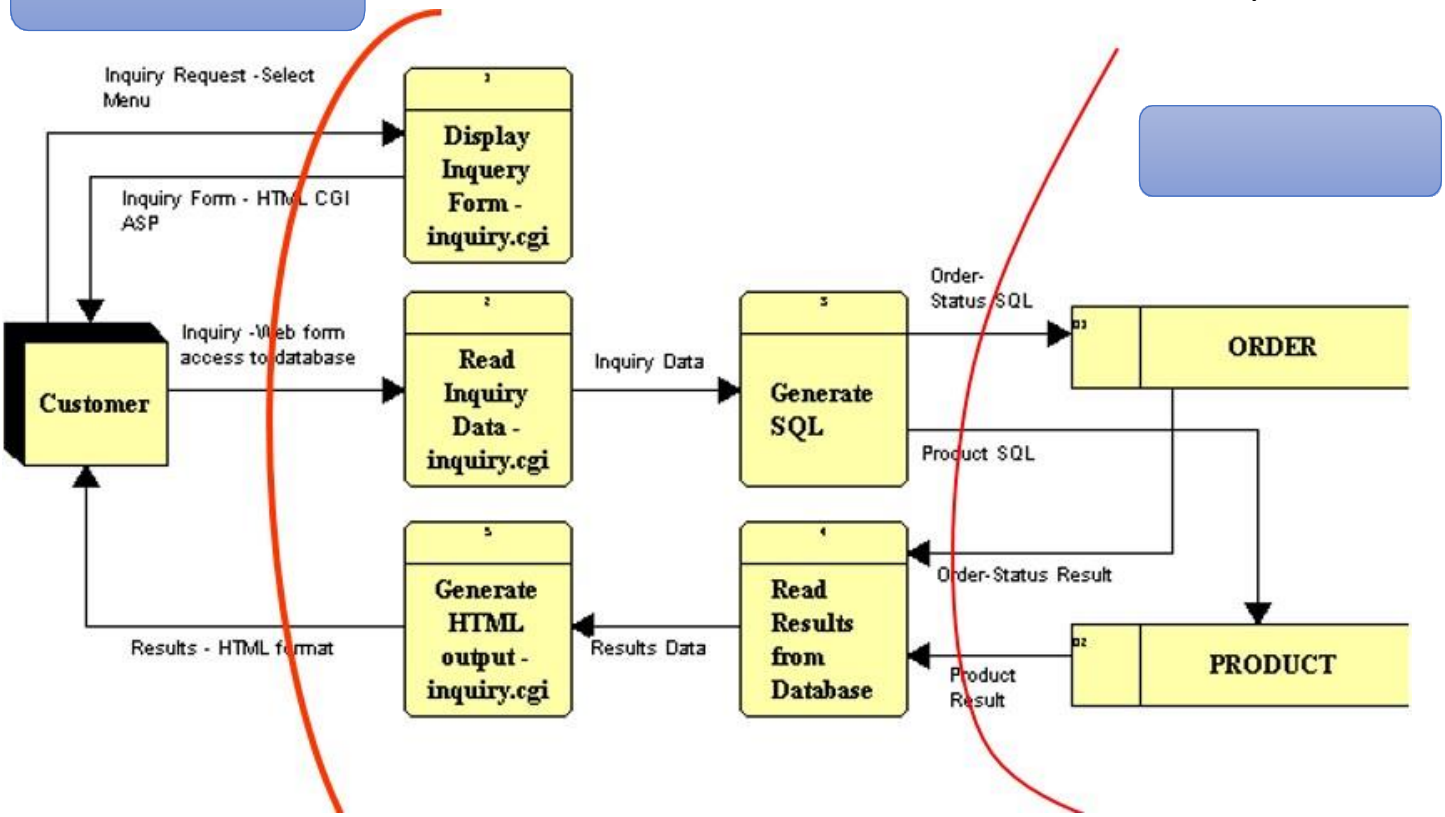
- To keep the screen simple by giving proper sequence, information, and clear captions.
- To meet the intended purpose by using appropriate forms.
- To ensure the completion of form with accuracy.
- To keep the forms attractive by using icons, inverse video, or blinking cursors etc.
- To facilitate navigation.

Form Driven Methodology

Input and Output Methods

Input Data

Output



Designing Effective Input

- ✓ The quality of system input determines the quality of system output.
- ✓ It is vital that input forms, displays, and interactive Web documents be designed with this critical relationship in mind.
- ✓ Well-designed input forms, interactive Web fill-in forms should meet the objectives of effectiveness, accuracy, ease of use, consistency, simplicity, and attractiveness.

Good Form Design

To design forms that people find useful, four guidelines for form design should be observed:

1. Make forms easy to fill in.
2. Ensure that forms meet the purpose for which they are designed.
3. Design forms to ensure accurate completion.
4. Keep forms attractive.

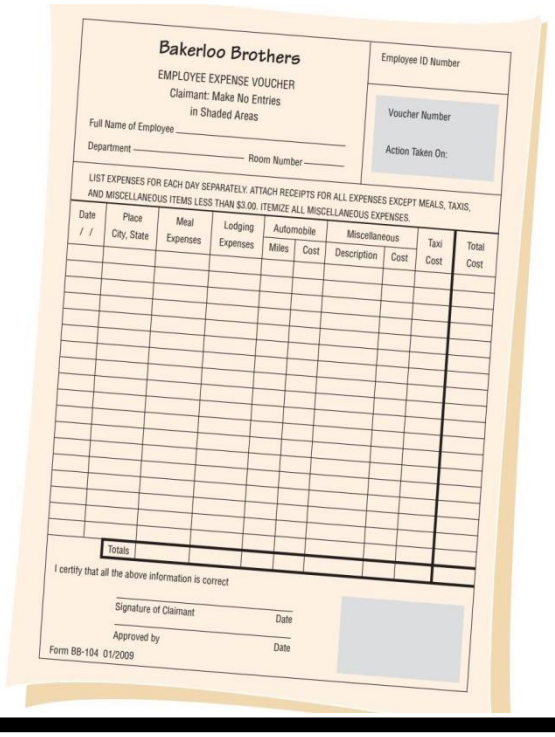
Making Forms Easy to Fill in: Form Flow

- ✓ Designing a form with proper flow can minimize the time and effort expended by employees in form completion.
- ✓ Forms should flow from left to right and top to bottom. Illogical flow takes extra time and is frustrating.
- ✓ A form that requires people to go directly to the bottom of the form and then skip back up to the top completion exhibits poor flow

Making Forms Easy to Fill in: Seven Sect.

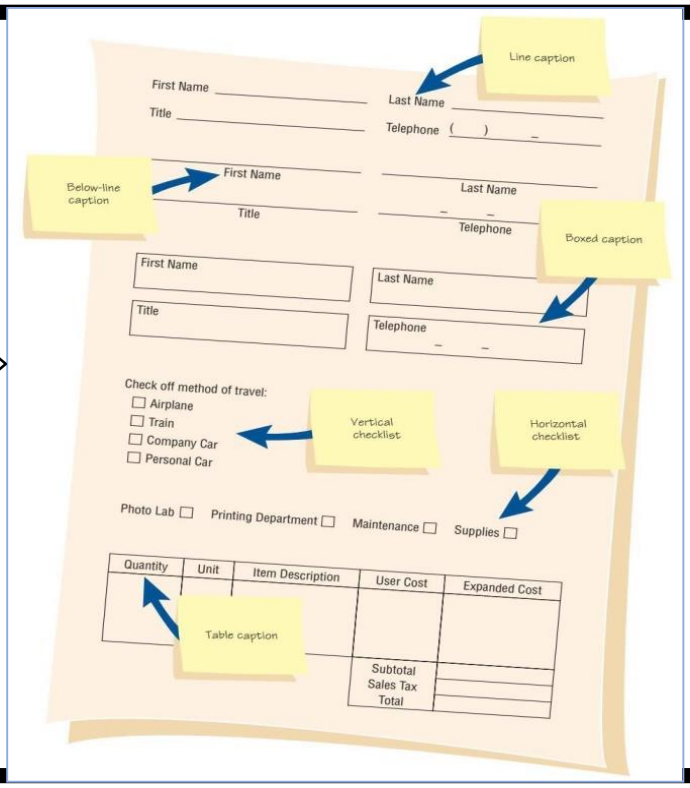
A second method that makes it easy for people to fill out forms correctly is logical grouping of information. The seven main sections of a form are the following:

1. Heading
2. Identification and access
3. Instructions
4. Body
5. Signature
6. Totals
7. Comments



Making Forms Easy to Fill in: Seven Sect

Major Captioning Alternatives



Data Input Methods

- ✓ **ON-LINE** — User directly enters data using screen prompts.
- ✓ **OFF-LINE** — Forms filled by users.

  Example- candidates for admission to a college fill forms

- ✓ Data from forms keyed in by a data entryoperator

Good design and Bad Design

Name

Address

Tick as applicable

Individual
Hindu undivided family
Parent/Guardian of minor

Bad design : Tendency will be to fill name on top line. Not enough space for letters of address

Bad design : Choices are not codified. Data entry operator will be confused.

Good design and Bad Design

Enter date

Day		month		Year	

(Good design)

Enter date

(Bad design)

Enter time

Hr		Min		Sec	

(Good design)

Enter time

(Bad design)

Good design and Bad Design

- **MENUS -** User presented several alternatives and asked to type his/her choice

Example: SELECT ALTERNATIVE

Type 1 For entering new student record

Type 2 For deleting student record

Type 3 For changing student record

Good design and Bad Design

Example

Roll no

Name FIRST NAME/INTIALS LAST NAME

Dept code

CE	CS	ME	EE	IT

CODES

Year

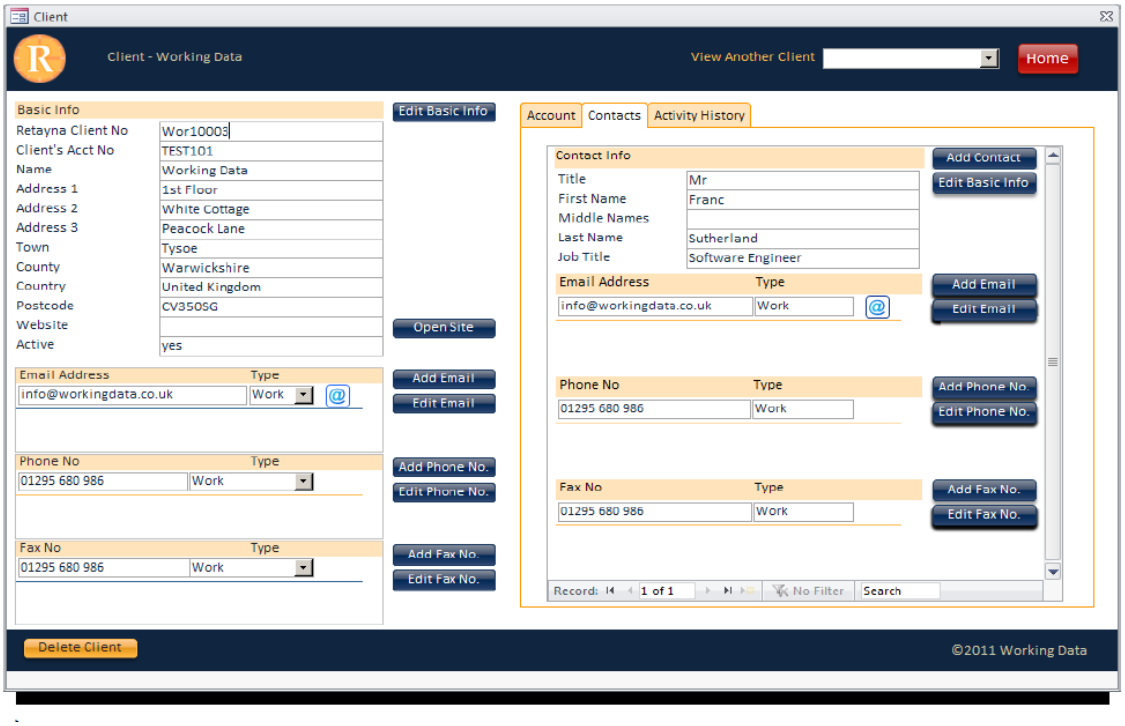
Hostel code

A	B	C	D

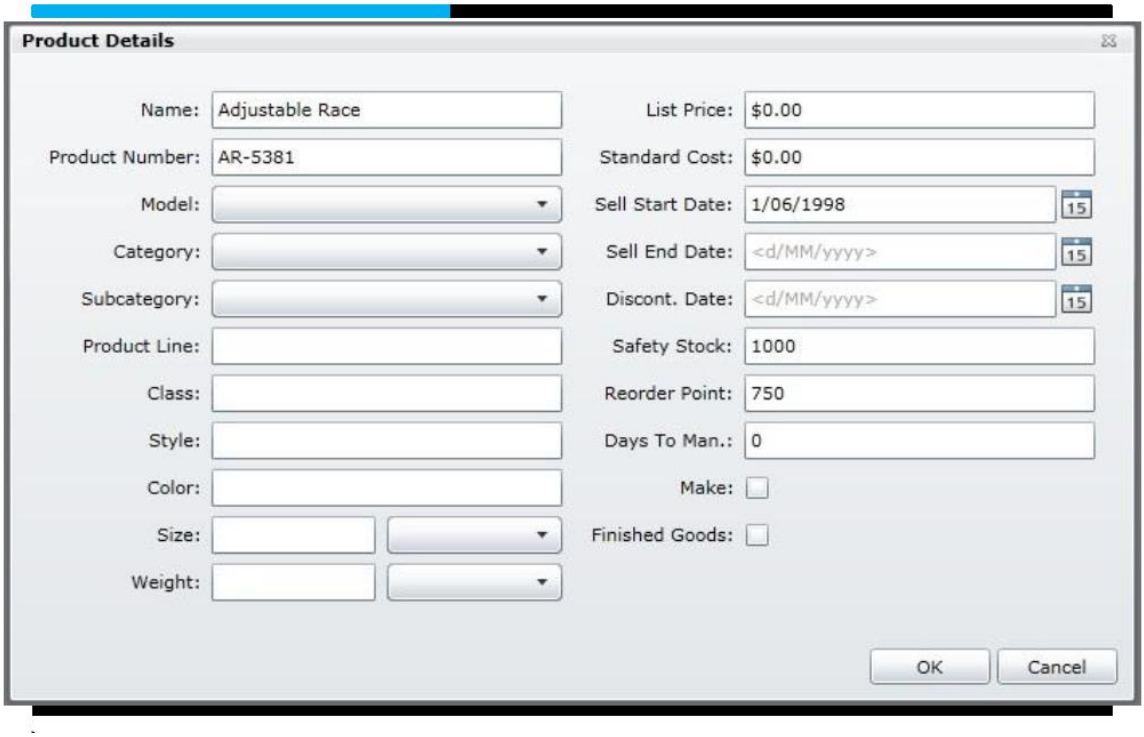
Pre-programmed to reject incorrect Roll no,Dept code,Year, Hostel code

- ✓ Has features to reject incorrect data input using built-in program.
- ✓ User friendly visual presentation.

Client Information Input Form



Product Details Input Form



Database Design

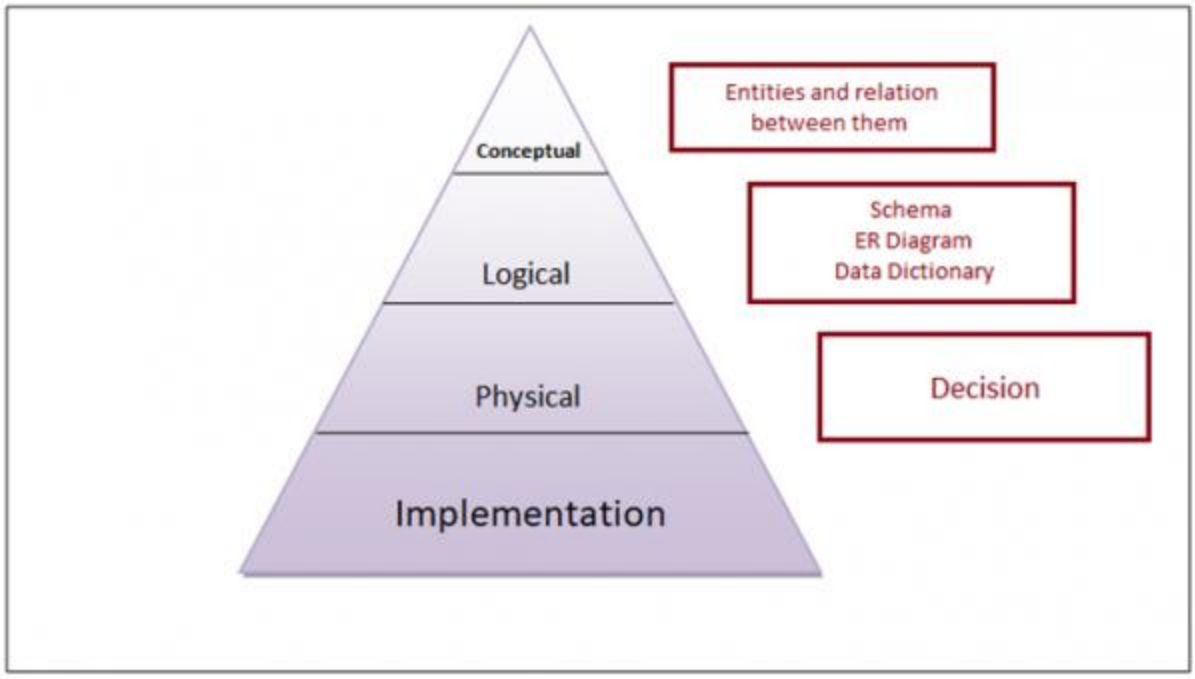
A database is a collection of information that is organized so that it can be easily accessed, managed and updated.

Data modeling

- Data modeling is often the first step in database design as the designers first create a

conceptual model of how data items relate to each other.

- Data modeling involves a progression from conceptual model to logical model to physical schema.



Database Design

Three main phases in Database Design:

1. **Conceptual database design** - to build the conceptual representation of the database, which has the identification of the important entities, relationships, and attributes.

2. **Logical database design** - to convert the conceptual representation to the logical structure of the database, which includes designing the relations (ERD).

3. **Physical database design** - to decide how the logical structure is to be physically implemented (as base relations) in the target Database Management System (DBMS).

Conceptual Data Modeling

- A conceptual data model identifies the highest-level relationships between the different entities.
- Designed and developed to be independent of DBMS, data storage locations or technologies.

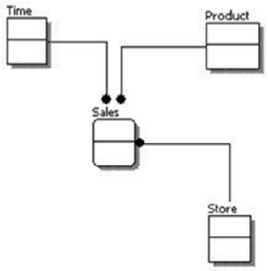
Features of conceptual data model include:

• Important entities and the relationships among them.

• No attribute is specified.

• No primary key is specified.

The figure below is an example of a conceptual data model.



From the figure above, we can see that the only information shown via the conceptual data model is the entities that describe the data and the relationships between those entities. No other information is shown through the conceptual data model.

Logical Data Modeling

A logical data model is a fully-attributed data model that is independent of DBMS, technology, data storage or organizational constraints. It typically describes data requirements from the business point of view.

A logical data model describes the data in as much detail as possible.

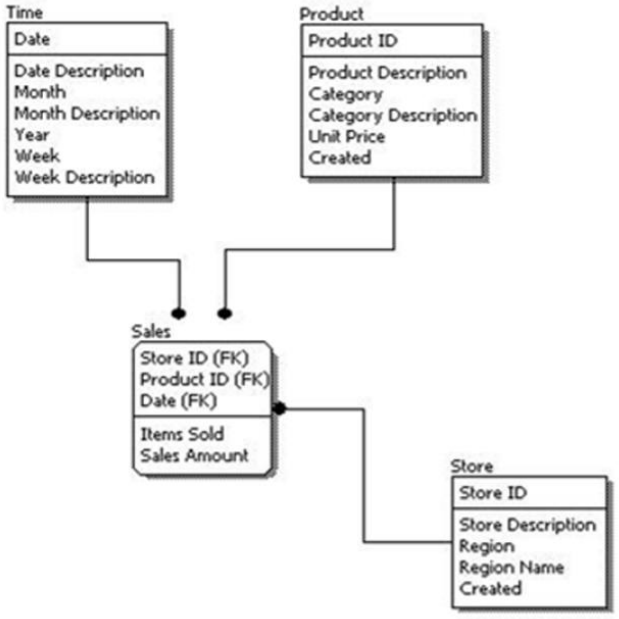
Features of a logical data model include:

- All entities and relationships among them.
- All attributes for each entity are specified.
- The primary key for each entity is specified.
- Foreign keys (keys identifying the relationship between different entities) are specified.
- Normalization occurs at this level.

Logical Model Design

The steps for designing the logical data model are as follows:

1. Specify primary keys for all entities.
2. Find the relationships between different entities.
3. Find all attributes for each entity.
4. Resolve many-to-many relationships.
5. Normalization.



Physical data model

A physical data model is a fully-attributed data model that is dependent upon a specific version of a data persistence technology.

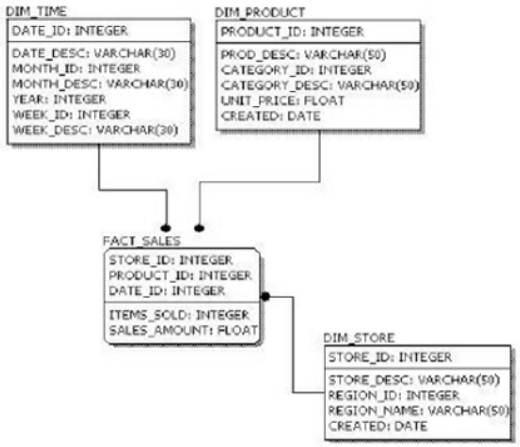
- The target implementation technology may be a relational DBMS, an XML document, a NoSQL data storage component, a spreadsheet or any other data implementation option.
- ***Features of a physical data model include:***

- ✓ Specification all tables and columns.
- ✓ Foreign keys are used to identify relationships between tables.
- ✓ Denormalization may occur based on user requirements.
- ✓ Physical considerations may cause the physical data model to be quite different from the logical data model.
- ✓ Physical data model will be different for different RDBMS. For example, data type for a column may be different between Oracle, DB2 etc.

Physical data model

The steps for physical data model design are as follows:

1. Convert entities into tables.
2. Convert relationships into foreign keys.
3. Convert attributes into columns.
4. Modify the physical data model based on physical constraints / requirements.



Data Models

Here we compare these three types of data models. The table below compares the different features:

Feature	Conceptual	Logical	Physical
Entity Names	✓	✓	
Entity Relationships	✓	✓	
Attributes		✓	
Primary Keys		✓	✓
Foreign Keys		✓	✓
Table Names			✓
Column Names			✓
Column Data Types			✓

Introduction to Entity-Relationship (E-R) Modeling

Entity-Relationship (E-R) Diagram

– A detailed, logical representation of the entities, associations and data elements for an organization or business

• Notation uses three main constructs

– Data entities

– Relationships

– Attributes

Entities

• An entity is a person, place, object, event or concept about which an organization wishes to maintain data.

• **Some Examples**: EMPLOYEE, STUDENT,STORE, STATE, MACHINE, SALE, ACCOUNT

• **Entity type** (or entity class) collection of entities with common characteristics (Eg:STUDENT)

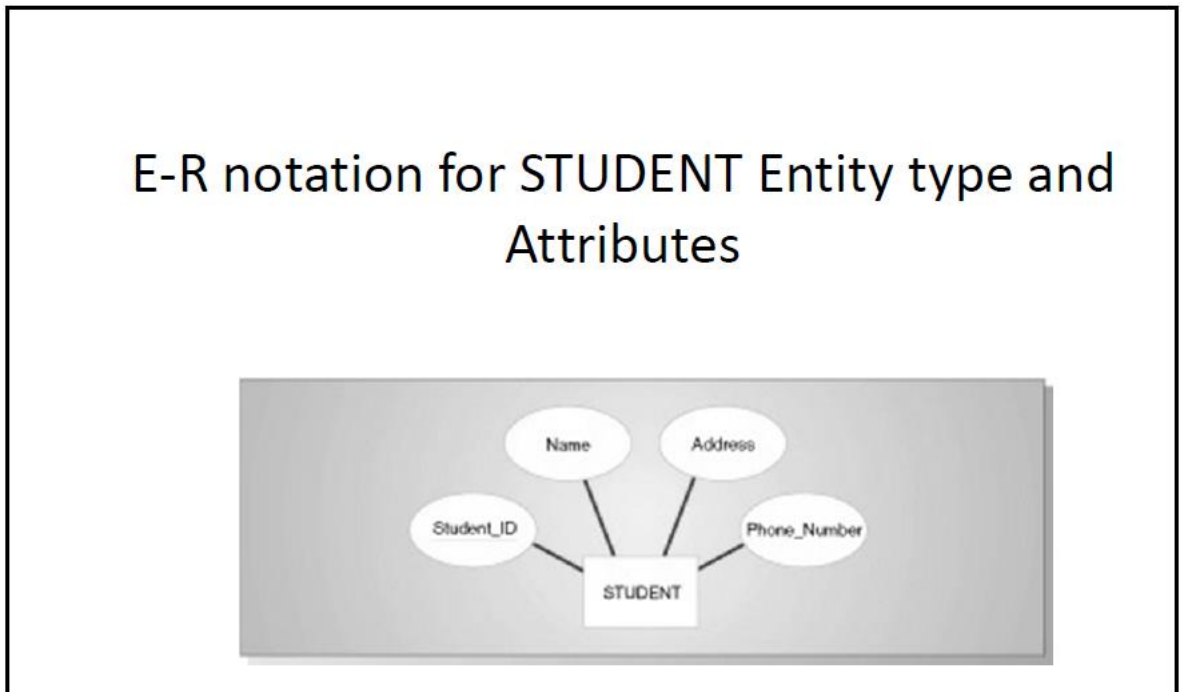
• **Entity instance**: A single occurrence of an entity type (E.g Eugene Ching)

• We use capital letters in naming an entity type and in an ER diagram, the name is placed inside a rectangle representing the entity.

Attributes

Each entity type has a set of attributes associated with it. An attribute is a property or characteristic of an entity that is of interest to the organization.

• **Example:** STUDENT: Student_ID, Student_Name, Home_Address, Phone_Number, Major



Primary Key

A primary key is a column (or columns) in a table that uniquely identifies the rows in that table.

Customers:

CustomerNo	FirstName	LastName
1	Sally	Thompson
2	Sally	Henderson
3	Harry	Henderson
4	Sandra	Wellington

For example, in the table above, CustomerNo is the primary key.

- ✓ The values placed in primary key columns must be unique for each row: no duplicates can be tolerated.
- ✓ In addition, nulls are not allowed in primary key columns.

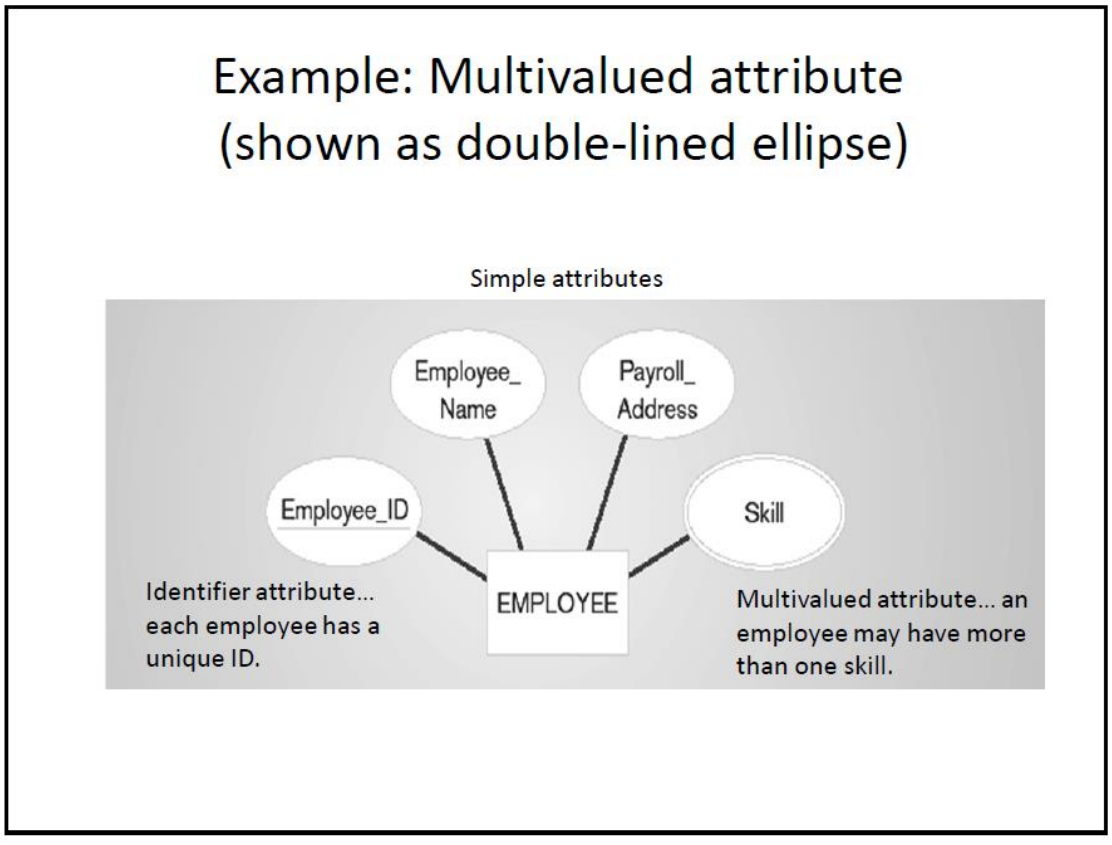
Multivalued Attributes

An attribute that may take on more than one value for each entity instance

• Represented on E-R Diagram in two ways:

– double-lined ellipse

– weak entity



Relationships

Relationship

– An association between the instances of one or more entity types that is of interest to the organization

– Association indicates that an event has occurred or that there is a natural link between entity types

– Relationships are always labeled with verb phrases

Degree of Relationship

- Degree: number of entity types that participate in a relationship
- Three cases
  - **Unary:** between two instances of one entity type
  - **Binary:** between the instances of two entity types
  - **Ternary:** among the instances of three entity types

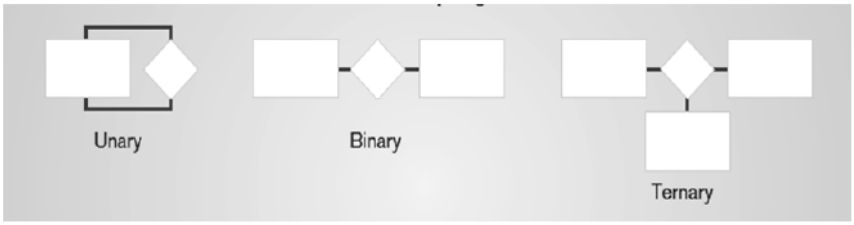
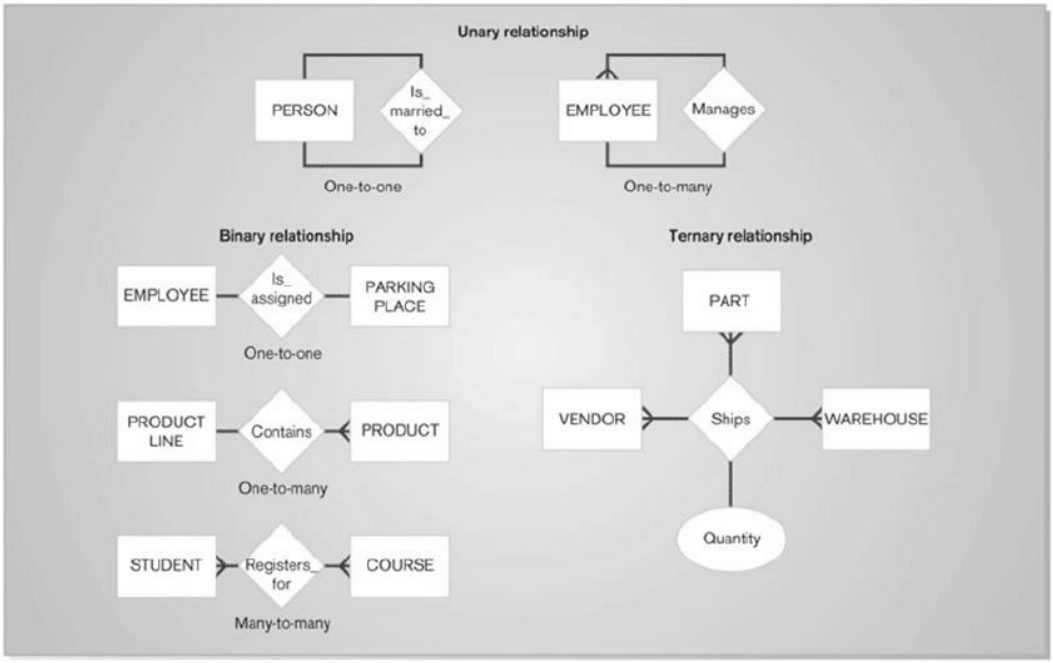


Figure 9-6 Example relationships of different degrees



Cardinality

The number of instances of entity B that can or must be associated with each instance of entity A

• **Minimum Cardinality**

– The minimum number of instances of entity B that may be associated with each instance of entity A

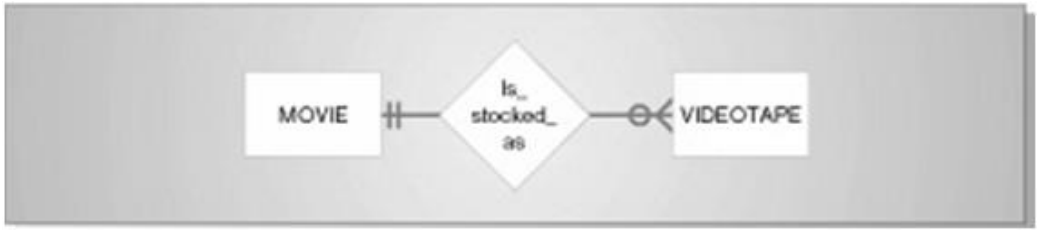
• **Maximum Cardinality**

– The maximum number of instances of entity B that may be associated with each instance of entity A

• **Mandatory vs. Optional Cardinalities**

– Specifies whether an instance must exist or can be absent in the relationship

Cardinality Example



ERD Diagram Design

Problem-1

- Drawing of ER model of university database application considering the constraints –
  - ✓ A university has many departments.
  - ✓ Each department has multiple instructors (one person is HOD). Here the HOD refers to the head of department.
  - ✓ An instructor belongs to only one department.
  - ✓ Each department offers multiple courses, each subject is taught by a single instructor.
  - ✓ A student may enroll for many courses offered by different departments.

Solution

Follow the steps given below to draw an Entity Relationship (ER) diagram for a University database application −

Step 1 – Identifying the entity sets.

The entity set has multiple instances in a given business scenario.

As per the given constraints the entity sets are as follows −

- ✓ Department
- ✓ Course
- ✓ Student
- ✓ Instructor

Head of the Department (HOD) is not an entity set. It is a relationship between the instructor and department entities.

Step 2 – Identifying the attributes for the given entities

- **Department** − the relevant attributes are department Name and location.
- **Course** − The relevant attributes are courseNo, course Name, Duration, and prerequisite.
- **Instructor** − The relevant attributes are Instructor Name, Room No, and telephone number.
- **Student** − The relevant attributes are Student No, Student Name, and date of birth

Step 3 – Identifying the Key attributes

- Department Name is the key attribute for Department.
- CourseNo is the key attribute for Course entity.
- Instructor Name is the key attribute for the Instructor entity.
- StudentNo is the key attribute for Student entities.

Step 4 – Identifying the relationship between entity sets

The department offers multiple courses and each course belongs to only one department, hence cardinality between department and course if one to many.



One course is enrolled by multiple students and one student for multiple courses. Hence, relationships are many to many.



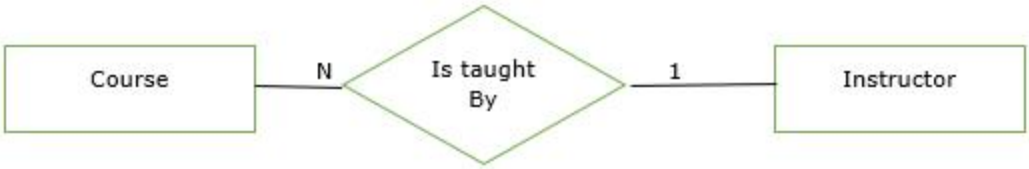
One department has multiple instructors and one instructor belongs to one and only one department, hence the relationship is one to many.



Each department has one “HOD” and one instructor is “HOD” for only one department, hence the relationship is one to one. Here, HOD refers to the head of the department.



One course is taught by only one instructor but one instructor teaches many courses hence the relationship between course and instructor is many to one.

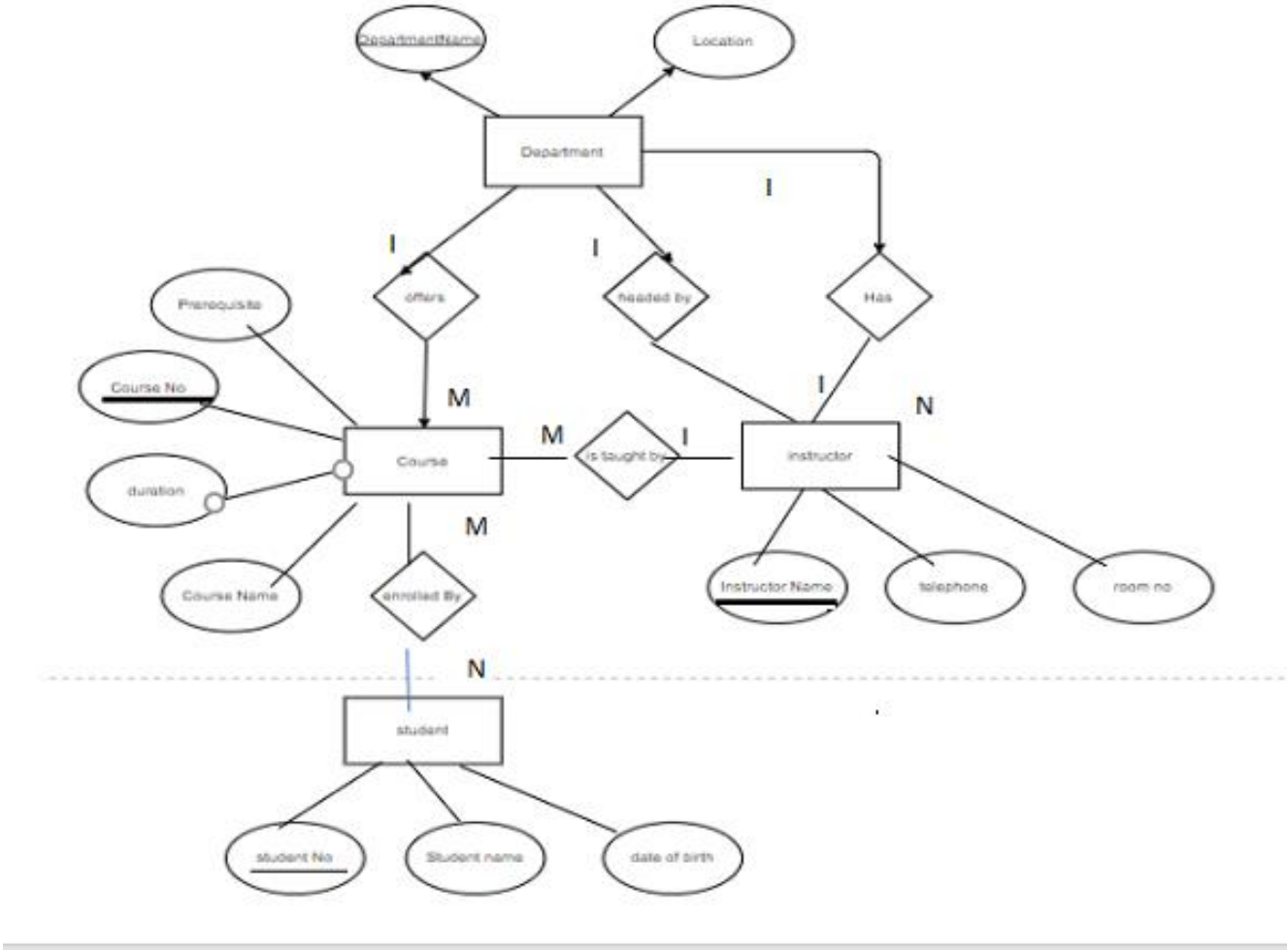


The relationship between instructor and student is not defined because of the following reasons −

- There is no significance in the relationship.
- We can always derive this relationship indirectly through course and instructors, and course and student.

Step 5 – Complete ER model

The complete ER Model is as follows −

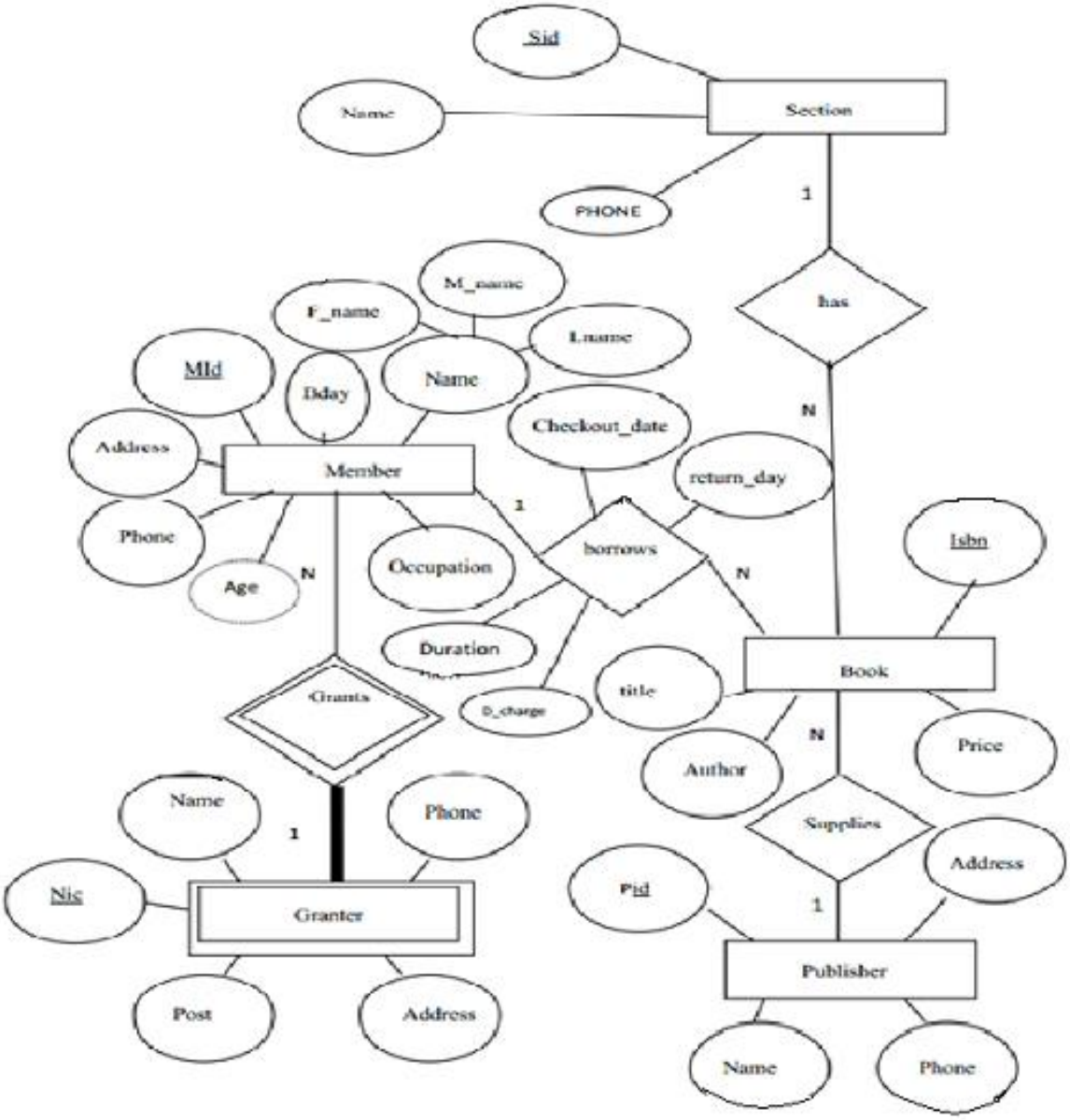


Problem-2

Draw an ER model for library management application considering the following constraints −

- ✓ In a library multiple students can enroll.
- ✓ Students can become a member by paying an appropriate fee.
- ✓ The books in the library are identified by a unique ID.
- ✓ Students can borrow multiple books from subscribed libraries.

Solution:



Problem-3: E-Commerce Site ERD

